Quantitative Investing Through Behavioral Finance

A Backtest of Momentum and Mean Reversion in Indian Large-Cap Equities

Indian Market Observatory

Contents

Abstract 3
1. Introduction 3
2. Theoretical Framework 4
    2.1 Efficient Market Hypothesis 4
    2.2 Behavioral Finance 5
    2.3 Momentum 5
    2.4 Mean Reversion 5
    2.5 Behavioral Explanations for Momentum 5
    2.6 Behavioral Explanations for Mean Reversion 6
    2.7 Research Gap and Motivation for This Backtest 6
3. Research Question and Hypotheses 6
4. Data 7
5. Methodology 8
    5.1 Monthly Return Calculation 8
    5.2 Momentum Signal 9
    5.3 Mean-Reversion Signal 9
    5.4 Stock Ranking 9
    5.5 Portfolio Selection 9
    5.6 Equal Weighting 9
    5.7 Monthly Rebalancing 9
    5.8 Portfolio Return 10

5.9 Portfolio Value 10
5.10 Benchmark Construction 10
5.11 Risk Metrics 10
5.12 Sharpe Ratio 10
5.13 Maximum Drawdown 10
5.14 Look-Ahead Bias Prevention 11
5.15 Transaction-Cost Assumptions 11
6. Baseline Results 11
6.1 Baseline Portfolio Performance 11
6.2 Cumulative Portfolio Value 12
6.3 Total Returns 13
6.4 CAGR 13
6.5 Volatility 13
6.6 Sharpe Ratio 13
6.7 Maximum Drawdown 14
6.8 Best and Worst Months 14
6.9 Positive and Negative Months 14
6.10 Monthly Strategy Behavior 14
6.11 Key Findings 15
7. Robustness and Sensitivity 15
7.1 Common Active-Period Robustness Check 15
7.2 Transaction-Cost Sensitivity 16
7.3 Concentration 17
7.4 Benchmark Limitation 17
8. Behavioral Finance Framework 17
8.1 Momentum and Possible Underreaction 18
8.2 Mean Reversion and Possible Overreaction 18
8.3 A Necessary Caution 18
9. Limitations 18
10. Future Research 19
11. Conclusion 20
12. References 20
13. Appendix 21
Appendix A: Formulas 21
Appendix B: Monthly Stock Selections (Momentum) 21
Appendix C: Monthly Stock Selections (Mean Reversion) 22
Appendix D: Robustness Results 23
Appendix E: Reproducibility and Data Integrity 23

Abstract

This paper tests whether two simple, rule-based trading strategies, momentum and mean reversion, can outperform a passive benchmark in a small sample of Indian large-cap stocks. Using monthly adjusted closing prices for five companies listed on the National Stock Exchange of India (Reliance Industries, Tata Consultancy Services, HDFC Bank, Infosys, and ICICI Bank) from August 2023 to June 2026, this study builds two rule-based portfolios. Each month, the momentum strategy buys the two stocks with the highest return in the previous month, while the mean-reversion strategy buys the two stocks with the lowest return in the previous month. Both portfolios are equally weighted between the two stocks and rebalanced monthly. They are compared against the Nifty 50 Price Index, a passive, buy-and-hold benchmark that excludes dividends. Over this sample, the Nifty 50 Price Index grew from ₹100,000 to ₹123,953.45, while the momentum strategy reached ₹105,318.89 and the mean-reversion strategy finished at ₹99,876.35. Neither active strategy outperformed the benchmark, although momentum ended above its starting capital and mean reversion finished close to flat. The results are discussed alongside ideas from behavioral finance, connecting momentum to possible investor underreaction and mean reversion to possible investor overreaction, while being explicit that this backtest does not measure investor psychology directly. Two robustness checks, a common active-period comparison and a transaction-cost sensitivity test, are also reported. The study is limited by its small stock universe, short time span, and simplified cost assumptions, and its findings should not be read as investment advice or as evidence about how these strategies would perform in the future.

1. Introduction

Every investor, from a large pension fund to a first-year college student with a small brokerage account, eventually asks the same question: is there a simple rule that can beat just buying and holding the market? Quantitative investing tries to answer this question with data rather than intuition. Instead of picking stocks based on a hunch or a news headline, a quantitative strategy follows a fixed, mechanical rule, applies it consistently, and lets the historical record show what would have happened. This kind of testing is called a backtest, and it is the main method used in this paper.

One reason quantitative investing is interesting is that it forces discipline. A rule that says 'buy the two stocks that went up the most last month' can be checked, repeated, and criticized in a way that a gut feeling cannot. This does not make the rule correct, but it makes it honest: either the numbers support it over a given period, or they do not.

A second reason this project is interesting is the connection to investor behavior. Classical finance theory, built around the efficient market hypothesis, assumes that prices already reflect all available information, so there should be no easy pattern left to exploit (Fama, 1970). But a large body of research in behavioral finance argues that real investors are not perfectly rational. They can be slow to update their views, they can overreact to bad news, and they can follow the crowd (Shiller, 2003). Two of the most studied patterns that might come from this kind of behavior are momentum, where stocks that have recently done well keep doing well for a while, and mean reversion, where stocks that have recently done badly bounce back. This project does not try to prove which side of that

debate is correct. It simply tests whether two very simple versions of these two ideas would have produced different results from a passive index over one specific period in the Indian stock market.

Indian large-cap equities were chosen for a practical reason: they are some of the most widely followed, heavily traded companies in the Indian market, so their price data is reliable and easy to obtain through public sources. Reliance Industries, Tata Consultancy Services, HDFC Bank, Infosys, and ICICI Bank are five such companies, spanning energy, information technology, and banking. They are not meant to represent the entire Indian stock market, and the results in this paper should not be generalized beyond this specific five-stock sample.

The central research question of this project is: can simple systematic strategies based on momentum and mean reversion outperform passive investing in Indian large-cap equities? To test this, two portfolios were built using only information that would have been available at the time of each trade, month by month, from August 2023 to June 2026. Both were compared against the Nifty 50 Price Index, a standard measure of the Indian stock market that does not include dividends.

The short answer, previewed here and explained in full in Section 6, is that over this sample, neither strategy beat the benchmark. The Nifty 50 Price Index ended with more value than either the momentum or the mean-reversion portfolio. Momentum finished above its starting capital, while mean reversion finished close to where it started. These results do not mean that momentum or mean reversion never work, and they do not mean that active investing is pointless. They describe what happened in one small, specific test, under one specific set of rules, over one specific period of time.

The rest of this paper is organized as follows. Section 2 reviews the theory behind market efficiency, behavioral finance, momentum, and mean reversion. Section 3 states the research question and hypotheses. Section 4 describes the data. Section 5 explains the methodology in detail, including how look-ahead bias was avoided. Section 6 presents the results. Section 7 reports robustness checks. Section 8 discusses possible behavioral explanations for the findings. Sections 9 and 10 cover limitations and future research, and Section 11 concludes.

2. Theoretical Framework

2.1 Efficient Market Hypothesis

The efficient market hypothesis, most closely associated with Fama (1970), argues that stock prices quickly incorporate all available information. If this is true, a stock's past price movements should tell you nothing useful about where it is headed next, because everything knowable about the stock is already priced in. Under a strict version of this idea, there should be no simple rule based on past returns, like momentum or mean reversion, that consistently earns extra profit once risk is accounted for.

The efficient market hypothesis has been enormously influential, and it remains a reasonable starting assumption for how markets should behave. But it has also been challenged by decades of empirical research showing patterns in stock returns that are difficult to explain with the simplest version of the theory. This backtest is, in a small way, a direct test of that tension: it checks whether two of the

best-known return-based patterns actually show up, and if so, how large and how reliable they are, in one specific Indian sample.

2.2 Behavioral Finance

Behavioral finance is the study of how psychology affects financial decisions and, potentially, market prices. Kahneman and Tversky (1979) showed through their prospect theory that people do not evaluate gains and losses the way traditional economic theory assumes; losses tend to feel more painful than equivalent gains feel pleasant, and people often make decisions based on how a choice is framed rather than on the numbers alone. Shiller (2003) later argued that ignoring psychology when studying financial markets leaves out an important part of the picture, since real investors are not always the perfectly rational agents that classical models assume.

Behavioral finance does not claim that markets are random or that prices are meaningless. It argues that investor psychology can create temporary, exploitable patterns, and that these patterns can coexist with markets that are still mostly efficient most of the time.

2.3 Momentum

Momentum is the observation that stocks which have performed well recently tend to keep performing well for some period afterward, before the effect fades. Jegadeesh and Titman (1993) documented this pattern in U.S. stocks, showing that buying past winners and selling past losers produced significant returns over three-to-twelve-month holding periods, even after accounting for common measures of risk. Momentum has since been studied in many other markets and time periods, and it remains one of the most replicated patterns in empirical finance.

2.4 Mean Reversion

Mean reversion is, in a sense, the opposite idea: it suggests that unusually strong or unusually weak recent performance may not continue, and may instead move back toward a more typical level. De Bondt and Thaler (1985) found that portfolios of stocks that had performed very poorly over the previous three to five years went on to outperform portfolios of previous strong performers over the following few years, a pattern they linked to investor overreaction. It is worth noting that De Bondt and Thaler studied returns over multi-year horizons, while this project uses a much shorter, one-month signal; the mean-reversion strategy tested here is a simplified, short-horizon version of the broader idea, not a direct replication of their original study.

2.5 Behavioral Explanations for Momentum

Several theoretical models connect momentum to investor psychology. Hong and Stein (1999) proposed a model in which information about a company spreads gradually through the market rather than all at once, so prices underreact to news in the short run; investors who trade on trends can then profit from this gradual adjustment. Other researchers have pointed to herding, where investors copy each other's trades, and confirmation bias, where investors pay more attention to information that supports a view they already hold, as further possible contributors to momentum. These are theoretical mechanisms proposed in the academic literature, not something directly tested by this backtest.

2.6 Behavioral Explanations for Mean Reversion

Mean reversion is often linked to the opposite behavioral pattern: overreaction. Barberis, Shleifer, and Vishny (1998) built a model in which investors put too much weight on recent news, causing prices to swing further than the underlying fundamentals justify, before correcting later. Loss aversion, the tendency identified by Kahneman and Tversky (1979) for losses to feel worse than equivalent gains feel good, has also been suggested as a driver of panic selling after bad news, which could push a stock's price below where it would otherwise settle. Anchoring, where investors fixate on a recent price level, is another commonly cited mechanism. Again, these are possible explanations drawn from the wider literature, not conclusions drawn from this project's own data.

2.7 Research Gap and Motivation for This Backtest

Most of the classic momentum and mean-reversion studies were conducted on U.S. or other developed-market data, over long sample periods, using large numbers of stocks. There is comparatively little publicly available, fully transparent, reproducible testing of these same simple ideas on a small, specific sample of Indian large-cap stocks over a recent period. This project does not attempt to fill that gap in a rigorous academic sense, given its small scale, but it does apply the same basic logic (previous-month return ranking, equal weighting, and monthly rebalancing) to a concrete and current Indian dataset, and reports the results honestly, including where they do not support the strategies being tested.

3. Research Question and Hypotheses

Research question: Can simple systematic strategies based on momentum and mean reversion outperform passive investing in Indian large-cap equities?

To test this question in a structured way, three hypotheses were set before looking at the results:

H1: A simple momentum strategy based on previous-month returns will generate a different cumulative return from the Nifty 50 Price Index over the study period.

This hypothesis is tested because momentum is one of the two return-based patterns this project investigates, and the most direct way to test it is to compare its cumulative outcome against a passive alternative over the same period.

H2: A simple mean-reversion strategy based on selecting the two worst previous-month performers will generate a different cumulative return from the Nifty 50 Price Index over the study period.

This hypothesis mirrors H1 but for the mean-reversion rule, allowing the two strategies to be compared not only against the benchmark but against each other.

H3: The two strategies may display different risk characteristics from the benchmark.

Return is only one dimension of performance. H3 is included because a strategy could match or exceed the benchmark's return while taking on very different volatility or drawdown, which matters for anyone trying to judge whether a strategy's return is worth its risk.

None of these hypotheses assumes in advance that the strategies will outperform. They are framed as tests of whether a difference exists, not predictions that the difference will favor the active strategies.

4. Data

All price data for this project came from Yahoo Finance, accessed through the yfinance library, a widely used, free tool for downloading historical stock price data. The dataset covers five stocks: Reliance Industries (RELIANCE.NS), Tata Consultancy Services (TCS.NS), HDFC Bank (HDFCBANK.NS), Infosys (INFY.NS), and ICICI Bank (ICICIBANK.NS). All five are listed on the National Stock Exchange of India and are among the most widely held and heavily traded companies on that exchange.

These five companies were not selected because of any special statistical property. They form a small, convenient sample of large, widely followed Indian companies from different sectors (energy, information technology, and banking), chosen for their data reliability and public familiarity. They are not intended to represent the Indian stock market as a whole, and no claim is made that they are an optimal or representative universe.

The study period runs from August 2023 to June 2026, using monthly adjusted closing prices. Adjusted closing prices were used specifically so that stock splits, bonus issues, and similar corporate actions would not create artificial jumps in the calculated returns. This produced 35 monthly observations for each of the five stocks, for a total of 175 rows of monthly stock data.

The benchmark used throughout this paper is the Nifty 50 Price Index, tracked under the ticker ^NSEI, over the same August 2023 to June 2026 period, also with 35 monthly observations. It is important to state clearly that this is the Nifty 50 Price Index, not the Nifty 50 Total Return Index. The Price Index reflects only the change in the prices of its 50 constituent companies; it does not include the dividends those companies pay out. A Total Return Index would include reinvested dividends and would likely show a somewhat higher cumulative value over the same period. This distinction is discussed further as a limitation in Section 9.

Both portfolios and the benchmark started with an initial capital of ₹100,000, a round, easy-to-interpret figure with no special significance beyond making percentage changes easy to read as rupee amounts.

Before any strategy calculations were performed, the dataset was checked for common data problems. This included checking for missing values, duplicate dates, correct chronological ordering, and complete monthly coverage across the full study period for every stock and for the benchmark. No missing values, duplicate dates, or coverage gaps were found in the final dataset used for this paper. Table 1 summarizes the universe and data description, and Figure 1 shows how the five stocks moved over the study period.

Table 1. Universe and data description

Field	Detail
Data source	Yahoo Finance, via yfinance
Stocks	RELIANCE.NS, TCS.NS, HDFCBANK.NS, INFY.NS, ICICIBANK.NS
Benchmark	Nifty 50 Price Index (^NSEI)
Study period	August 2023 - June 2026
Frequency	Monthly
Stock observations	175 rows (35 months x 5 stocks)
Benchmark observations	35 monthly rows
Initial capital	₹100,000

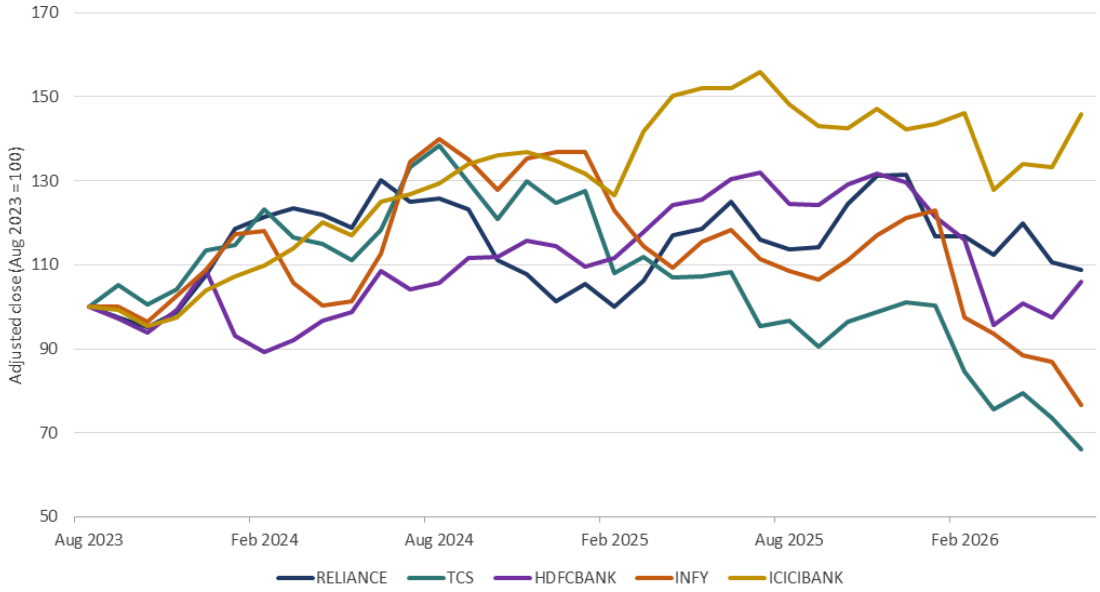


Figure 1. Adjusted monthly closing prices of the five sample stocks, August 2023 – June 2026 (August 2023 = 100)

Source: Yahoo Finance via yfinance (project dataset clean_monthly_stock_dataset.csv); index calculation by the author.

5. Methodology

5.1 Monthly Return Calculation

For every stock, the return for a given month was calculated from the adjusted closing price at the end of that month and the adjusted closing price at the end of the previous month:

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

where $P_{i,t}$ is stock i's adjusted closing price at the end of month t. This is a standard, simple percentage-change calculation applied consistently across every stock and every month in the sample.

5.2 Momentum Signal

The momentum signal for a stock is its own return during the previous month:

$$M_{i,t} = \frac{P_{i,t-1} - P_{i,t-2}}{P_{i,t-2}}$$

This is simply the stock's realized return one month before the holding month begins. No other information, such as longer-term trends or trading volume, was used in the signal.

5.3 Mean-Reversion Signal

The mean-reversion strategy uses the exact same signal, $M_{i,t}$, as momentum. The only difference between the two strategies is how the signal is used: momentum ranks stocks from highest to lowest signal and buys the top two, while mean reversion ranks stocks from lowest to highest signal and buys the bottom two. It is worth repeating clearly that this is a simple, return-based mean-reversion rule. It is not a z-score strategy, not a moving-average strategy, not a Bollinger-band strategy, and not a statistical distance-from-mean model. It is the plainest possible version of the idea: buy whatever did worst last month.

5.4 Stock Ranking

At the start of every month, all five stocks were ranked using their $M_{i,t}$ signal, calculated from the month before. Momentum selected the two highest-ranked stocks. Mean reversion selected the two lowest-ranked stocks.

5.5 Portfolio Selection

Each strategy therefore held exactly two of the five stocks at any given time. This is a concentrated construction: at any point, 60 percent of the five-stock universe was not held by either strategy. The consequences of this design choice are discussed in Section 7.3.

5.6 Equal Weighting

The two selected stocks always received equal weight, 50 percent each, rather than being weighted by their signal strength, market capitalization, or any other factor. Equal weighting was chosen for simplicity and transparency, in keeping with the goal of testing the plainest possible version of each strategy.

5.7 Monthly Rebalancing

At the end of each month, both strategies were rebalanced: the signal was recalculated using the month that had just ended, the five stocks were re-ranked, and the portfolio was updated to hold whichever two stocks the new ranking selected. If the same two stocks were selected again, the portfolio simply continued holding them.

5.8 Portfolio Return

For the two selected stocks in a given month, the portfolio's return was calculated as a simple weighted average:

$$R_{p,t} = 0.5 \cdot R_{1,t} + 0.5 \cdot R_{2,t}$$

where $R_{1,t}$ and $R_{2,t}$ are the realized returns of the two selected stocks during month t, the same month they were held.

5.9 Portfolio Value

Portfolio value was tracked by compounding this monthly return forward from the ₹100,000 starting capital:

$$V_t = V_{t-1} \cdot (1 + R_{p,t})$$

This was done separately for the momentum portfolio and the mean-reversion portfolio, each starting from the same ₹100,000 base.

5.10 Benchmark Construction

The Nifty 50 Price Index benchmark was constructed as a simple passive, buy-and-hold portfolio: ₹100,000 was treated as invested in the index at the start of the study period, and its value was compounded forward using the index's own monthly returns, with no rebalancing, no stock selection, and no trading of any kind.

5.11 Risk Metrics

Beyond raw return, this paper reports two standard risk measures: annualized volatility and maximum drawdown. Annualized volatility was calculated as the standard deviation of monthly returns, multiplied by the square root of twelve, converting a monthly measure of variability into a yearly one for easier comparison.

5.12 Sharpe Ratio

The Sharpe ratio, developed by Sharpe (1994) as a way to measure return relative to risk, was calculated using a risk-free rate assumption of zero percent. This is a simplification, not an estimate of the actual Indian risk-free rate; it was chosen so that the ratio would compare each portfolio's raw return to its own volatility, without introducing an unverified assumption about interest rates.

5.13 Maximum Drawdown

Maximum drawdown measures the largest peak-to-trough decline a portfolio experienced during the sample. At every point in time, the running maximum value the portfolio had reached up to that point was recorded, and the drawdown at that point was calculated as:

$$DD_t = \frac{V_t - RunningMax_t}{RunningMax_t}$$

The maximum drawdown reported for each portfolio is the most negative value this drawdown series ever reached during the study period.

5.14 Look-Ahead Bias Prevention

Look-ahead bias occurs when a backtest accidentally uses information that would not actually have been available at the time a decision was made, which can make a strategy look better than it really was. This project treated avoiding look-ahead bias as a core requirement, not an afterthought. The signal used to select month t's portfolio was always calculated using only information available at the end of month t-1: specifically, the return realized during month t-1 itself. The return actually earned during month t, used to measure how the portfolio performed, was only known after month t had ended, well after the selection had already been made.

Because this signal requires a full previous month of return data, the first two months of the sample, August 2023 and September 2023, did not have a valid signal available. August 2023 has no prior month in the dataset at all, and September 2023 would have needed August's return as its signal, which itself could not be calculated without a month before that. Both months are therefore excluded from active trading. October 2023 is the first month in which both strategies held a real, signal-based portfolio.

This rule was checked, not just assumed. Every stock and every month's signal was independently re-derived and compared against the realized return of the prior month, across the full sample. This check found zero mismatches: in every single case, the signal used to select a given month's portfolio was exactly equal to the realized return of the month before, with no case in which a future month's information leaked into an earlier decision.

5.15 Transaction-Cost Assumptions

The baseline results reported in Section 6 assume zero transaction costs and zero slippage, meaning trades are assumed to happen at the exact recorded price with no brokerage, taxes, or market-impact costs. This is a simplifying assumption, common in early-stage backtests, made explicit so it cannot be mistaken for a claim about real-world trading costs. Section 7.2 relaxes this assumption directly, testing how the results change under a range of illustrative transaction-cost levels.

6. Baseline Results

6.1 Baseline Portfolio Performance

Table 2 reports the full set of baseline performance metrics for the momentum strategy, the mean-reversion strategy, and the Nifty 50 Price Index benchmark, all measured over the same August 2023 to June 2026 sample, all starting from ₹100,000, and all using zero transaction costs.

Table 2. Baseline performance metrics, August 2023 – June 2026, zero transaction costs

Metric	Momentum	Mean Reversion	Nifty 50 Price Index
Final Value	₹105,318.89	₹99,876.35	₹123,953.45

Metric	Momentum	Mean Reversion	Nifty 50 Price Index
Total Return	+5.32%	-0.12%	+23.95%
CAGR	+1.85%	-0.04%	+7.87%
Annualized Volatility	18.75%	13.70%	13.94%
Sharpe Ratio (rf = 0%)	0.192	0.063	0.614
Maximum Drawdown	-22.06%	-19.78%	-14.78%
Best Month	+10.65% (Jun 2024)	+7.68% (Dec 2023)	+7.94% (Jun 2024)
Worst Month	-10.22% (Feb 2025)	-7.18% (Mar 2026)	-11.31% (Mar 2026)
Positive Months	18	16	19
Negative Months	15	17	15
Win Rate	54.5%	48.5%	55.9%

One correction is worth stating clearly here because it was flagged during the project's review process: an earlier draft of this project's supporting materials incorrectly described March 2026 as the worst month for all three portfolios. This is not accurate. As Table 2 shows, March 2026 was the worst month only for mean reversion (-7.18%) and the Nifty 50 Price Index (-11.31%). Momentum's worst month was February 2025, at -10.22%.

6.2 Cumulative Portfolio Value

Figure 2 shows the cumulative value of all three portfolios, all starting at ₹100,000 in August 2023 and ending in June 2026. The Nifty 50 Price Index climbs the most steadily and finishes clearly above both active strategies. The momentum portfolio shows larger swings, including a sharp rise during 2024 and a steep decline in the sample's final months, ending only modestly above its starting value. The mean-reversion portfolio moves in a narrower band than momentum for most of the sample and ends close to its starting value.

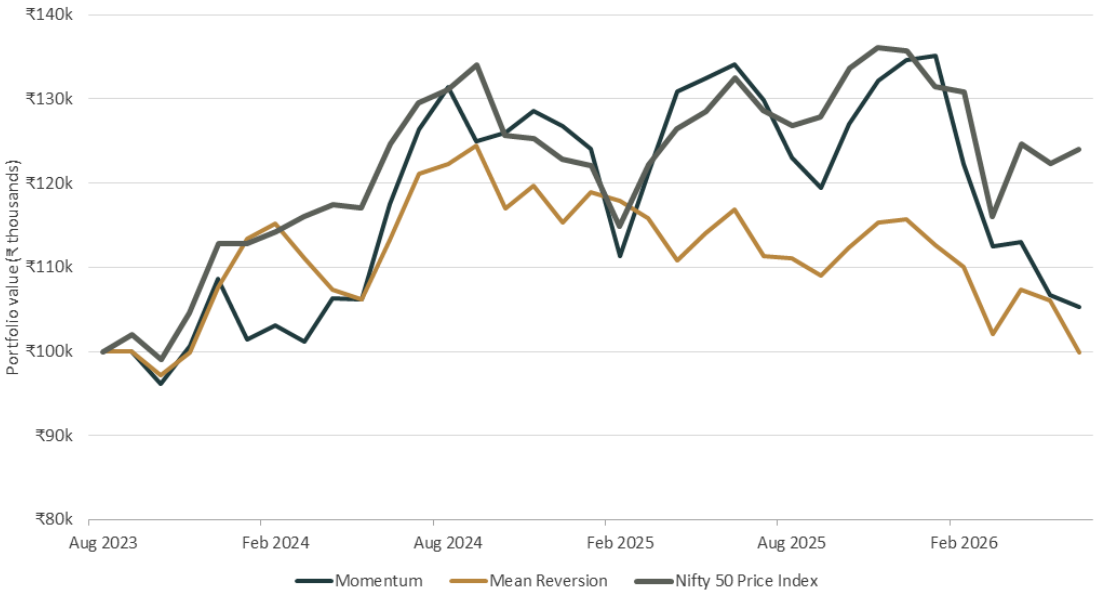


Figure 2. Cumulative value of ₹100,000 in each portfolio, August 2023 – June 2026 (zero transaction costs)

Source: Project backtest results (momentum_backtest.csv, mean_reversion_backtest.csv, nifty50_benchmark.csv).

6.3 Total Returns

Over the full sample, the Nifty 50 Price Index produced a total return of +23.95%, compared with +5.32% for momentum and -0.12% for mean reversion. In plain terms, ₹100,000 placed passively in the Nifty 50 Price Index grew to nearly ₹124,000, while the same amount run through the momentum rule grew to about ₹105,300, and the same amount run through the mean-reversion rule ended almost exactly where it started.

6.4 CAGR

The compound annual growth rate, or CAGR, restates total return as an equivalent constant yearly rate. The benchmark's CAGR of +7.87% is substantially higher than momentum's +1.85% and mean reversion's -0.04% over this roughly two-year-and-nine-month sample.

6.5 Volatility

Momentum was the most volatile of the three, at 18.75% annualized, compared with 13.70% for mean reversion and 13.94% for the benchmark. Mean reversion and the benchmark had similar volatility despite very different returns, which is itself a useful observation: momentum took on the most risk by this measure while also barely outperforming its own starting capital, whereas the benchmark achieved its higher return without correspondingly higher volatility.

6.6 Sharpe Ratio

The Sharpe ratio, which compares return to volatility, was 0.192 for momentum, 0.063 for mean reversion, and 0.614 for the Nifty 50 Price Index, all using a zero percent risk-free rate assumption.

On this measure, the benchmark delivered noticeably more return per unit of volatility than either active strategy over this sample.

6.7 Maximum Drawdown

Momentum's maximum drawdown of -22.06% was the steepest of the three, followed by mean reversion at -19.78%, and the benchmark at -14.78% (Figure 3). In other words, at its worst point during the sample, the momentum portfolio had lost about 22 percent of its peak value, a materially larger decline than either the mean-reversion portfolio or the passive benchmark experienced.

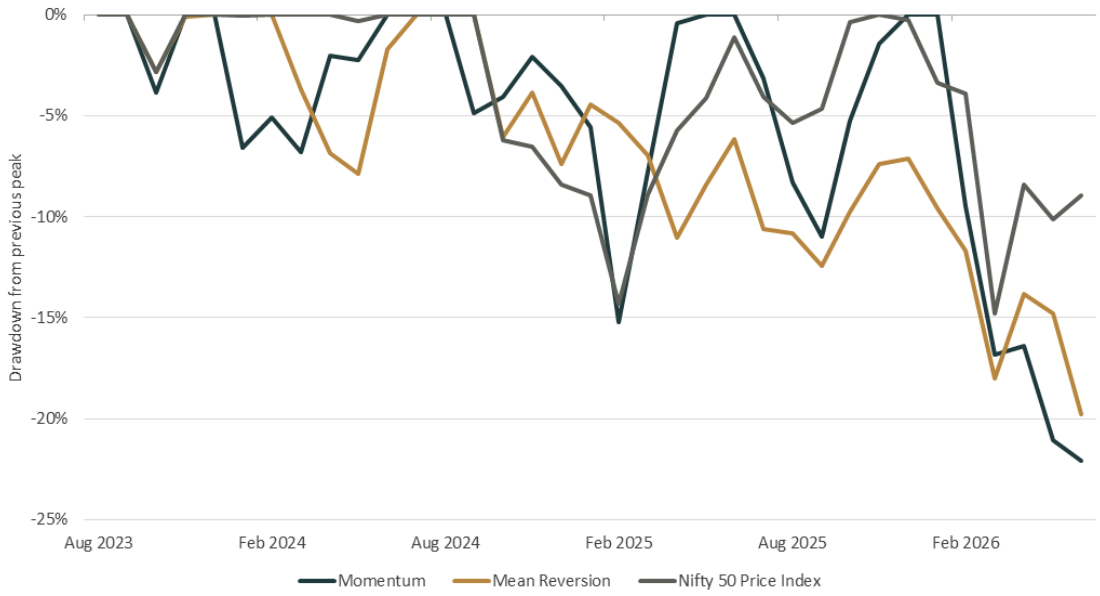


Figure 3. Drawdown from previous peak for each portfolio, August 2023 – June 2026

Source: Project backtest results (drawdown_momentum.csv, drawdown_mean_reversion.csv, drawdown_nifty50.csv).

6.8 Best and Worst Months

Momentum's best month was June 2024 at +10.65%, and its worst was February 2025 at -10.22%. Mean reversion's best month was December 2023 at +7.68%, and its worst was March 2026 at -7.18%. The benchmark's best month, also June 2024, was +7.94%, and its worst, March 2026, was -11.31%.

6.9 Positive and Negative Months

Momentum was positive in 18 of its 33 active months and negative in 15, for a win rate of 54.5%. Mean reversion was positive in 16 months and negative in 17, a win rate of 48.5%, meaning it was, by month count, negative slightly more often than it was positive. The benchmark was positive in 19 months and negative in 15, a win rate of 55.9%.

6.10 Monthly Strategy Behavior

Both strategies changed their two selected stocks frequently. Momentum rebalanced into a different pair of stocks in 28 of its 33 active months, and mean reversion did so in 27 of 33. This means that, in practice, both strategies were almost always trading, rarely holding the same two-stock

combination for two consecutive months. This high turnover is directly relevant to the transaction-cost sensitivity test in Section 7.2, since more frequent trading means more opportunities for real-world costs to erode returns.

Overall, the central empirical finding of this project is straightforward to state and important not to overstate: within this specific five-stock universe and this specific August 2023 to June 2026 sample, the Nifty 50 Price Index accumulated more value than either tested strategy. Momentum finished above its starting capital; mean reversion finished approximately flat. This does not prove that passive investing is always superior, and it does not prove that momentum or mean reversion never work; it describes what happened in this one test.

6.11 Key Findings

- The Nifty 50 Price Index finished at ₹123,953.45, compared with ₹105,318.89 for Momentum and ₹99,876.35 for Mean Reversion.
- Over this sample, the Nifty 50 Price Index accumulated more value than either tested active strategy.
- Momentum finished above its starting capital, while Mean Reversion finished approximately flat.

> *These findings describe this specific sample, universe, methodology, and cost assumption; they do not establish future performance or universal strategy effectiveness.*

7. Robustness and Sensitivity

A single backtest result, run under one specific set of assumptions, can be fragile. To check how sensitive the baseline findings are to some of the choices made in this project, two supplementary checks were run (Sections 7.1 and 7.2), followed by notes on concentration and the benchmark (Sections 7.3 and 7.4). None of these checks replace the baseline results in Section 6; they exist to show how much, or how little, the picture changes under different reasonable assumptions.

7.1 Common Active-Period Robustness Check

The baseline results in Section 6 start all three series in August 2023, but the active strategies do not have a valid trading signal until October 2023, so their first two months are simply flat at ₹100,000 while the benchmark is already moving. To remove this asymmetry, all three series were rebased to ₹100,000 starting in October 2023, the first month all three are genuinely comparable, and re-run through June 2026. Table 3 reports the results.

Table 3. Common active-period robustness check, rebased to ₹100,000 at October 2023

Metric	Momentum	Mean Reversion	Nifty 50 Price Index
Final Value	₹109,547.54	₹102,805.15	₹125,085.17
Total Return	+9.55%	+2.81%	+25.09%
CAGR	+3.48%	+1.04%	+8.76%
Volatility	18.86%	13.79%	14.19%

Metric	Momentum	Mean Reversion	Nifty 50 Price Index
Sharpe Ratio	0.273	0.142	0.663
Maximum Drawdown	-22.06%	-19.78%	-14.78%

The pattern from the baseline holds under this adjustment: the benchmark still ends with the highest value, return, and Sharpe ratio, and momentum still finishes ahead of mean reversion. The main effect of this check is that all three figures rise somewhat once the two 'no signal' months are removed, since the benchmark's early growth during August and September 2023 is no longer counted for it alone.

7.2 Transaction-Cost Sensitivity

The baseline assumes zero transaction costs, which is unrealistic for any real trading strategy. To test how sensitive the results are to this assumption, a round-trip transaction cost was applied to the active strategies whenever they rebalanced into a different stock, at four illustrative levels: 0, 10, 25, and 50 basis points. No cost was applied to the passive benchmark, since it never trades. None of these cost levels are claimed to be the single realistic cost for Indian equity trading; they are stress points meant to show a range of outcomes (Table 4 and Figure 4).

Table 4. Transaction-cost sensitivity, applied only to active strategies on rebalance months

Round-Trip Cost	Momentum Final Value	Mean Reversion Final Value
0 bps	₹105,318.89	₹99,876.35
10 bps	₹103,279.62	₹97,979.77
25 bps	₹100,291.61	₹95,199.47
50 bps	₹95,495.09	₹90,732.84

Because momentum rebalanced in 28 of 33 active months and mean reversion in 27 of 33, both strategies were exposed to cost on nearly every month of the sample. At 25 basis points, momentum's already modest gain is almost entirely erased, ending close to its ₹100,000 starting point. At 50 basis points, both strategies show a net loss on their starting capital. This test does not prove that 25 or 50 basis points is the correct cost assumption for a real Indian brokerage account; it shows that the baseline result for these two strategies is close enough to break-even that incorporating non-zero trading costs could materially reduce or eliminate the small gain observed for momentum.

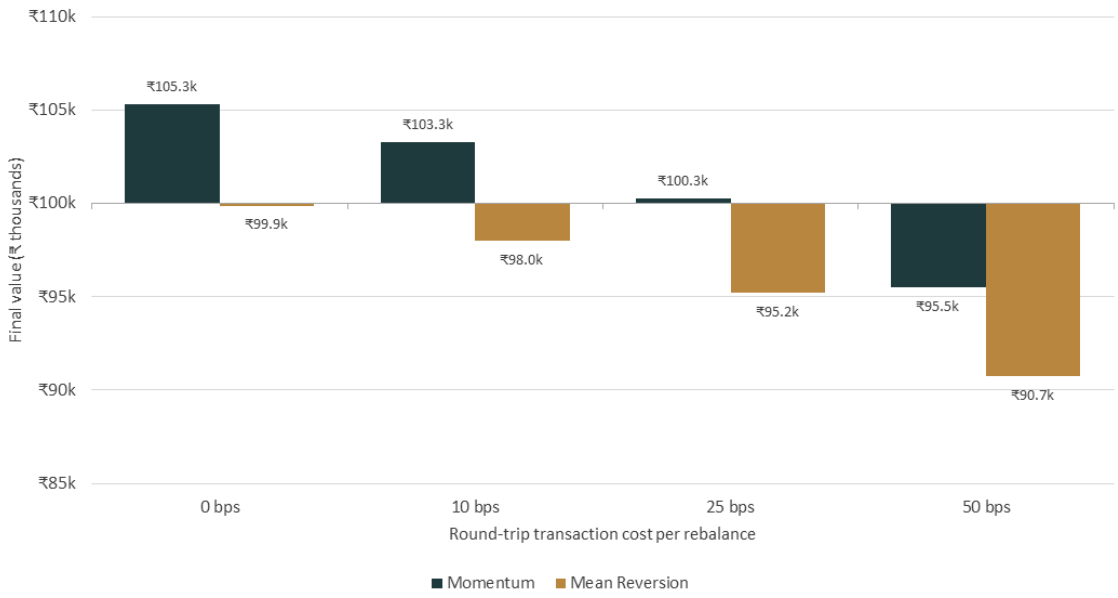


Figure 4. Final portfolio value under different round-trip transaction costs (bars start at the ₹100,000 starting capital)

Source: Project robustness results (transaction_cost_sensitivity.csv).

7.3 Concentration

Both strategies hold only two of the five available stocks at any given time. This is a concentrated construction: at every point in the sample, 60 percent of the universe was not held by either active strategy. This concentration may contribute to the higher volatility and drawdowns observed in the active strategies relative to the more diversified benchmark; however, causality was not experimentally isolated in this project. No side-by-side test was run comparing this two-stock construction against, for example, a three-stock or four-stock version of the same rules, so this remains a plausible explanation rather than a demonstrated cause.

7.4 Benchmark Limitation

As stated in Section 4, the benchmark used throughout this paper is the Nifty 50 Price Index, which excludes dividends. A Nifty 50 Total Return Index comparison, which would include reinvested dividends, would be a natural and useful future extension of this work, and it would likely show the benchmark performing somewhat better than reported here, since Indian large-cap companies typically pay some dividend. This paper does not attempt to estimate or insert a specific dividend yield, since doing so without real total-return data would introduce an unverified assumption into the results.

8. Behavioral Finance Framework

This project connects its two tested strategies to ideas from behavioral finance, but it is important to be precise about what this backtest can and cannot show. There is no investor-level survey data in this project, no measure of trader sentiment, and no direct observation of anyone's psychology. The connections drawn in this section are theoretical interpretations linking the mechanical rules tested

here to explanations proposed elsewhere in the academic literature, not causal findings established by this project's own data.

8.1 Momentum and Possible Underreaction

Momentum's underlying logic, that recent winners keep winning for a while, is conceptually consistent with several ideas discussed in Section 2.5. One possible explanation is investor underreaction: if new information about a company spreads gradually rather than all at once, as in the model proposed by Hong and Stein (1999), prices may take time to fully reflect good or bad news, allowing a trend to persist briefly. Herding, where investors follow other investors' trades rather than forming independent judgments, and confirmation bias, where investors pay more attention to information supporting a trend they already believe in, are further theoretical mechanisms sometimes proposed alongside underreaction. This backtest does not test any of these mechanisms directly; it only tests whether a return-based rule built on the assumption that trends persist would have made money in this specific sample, which, over this specific sample, it barely did before considering trading costs.

8.2 Mean Reversion and Possible Overreaction

Mean reversion's underlying logic, that recent losers may bounce back, is conceptually consistent with the idea of investor overreaction described by De Bondt and Thaler (1985) and modeled by Barberis, Shleifer, and Vishny (1998). One possible explanation is that investors overweight bad news, pushing a stock's price down further than its fundamentals justify, creating room for a partial recovery. Loss aversion, as described by Kahneman and Tversky (1979), and panic selling are sometimes proposed as behavioral drivers of this kind of overreaction, along with anchoring to a recent price level. Again, none of these mechanisms were tested directly here; the mean-reversion strategy simply tested whether buying last month's two worst performers produced a different outcome from the benchmark, and in this sample, it finished close to flat rather than clearly recovering.

8.3 A Necessary Caution

The results of this backtest do not establish that investor psychology caused the returns observed. It would be inaccurate, for example, to say that momentum worked because investors underreacted; the accurate statement is that the momentum result is broadly consistent with theories of underreaction, although this backtest does not directly test investor behavior. The same caution applies to mean reversion. These behavioral frameworks are offered as context for why researchers have historically been interested in momentum and mean reversion, not as an explanation proven by this project's own results.

9. Limitations

This project has several limitations that matter for interpreting its results honestly.

Small sample: the universe includes only five stocks, a tiny fraction of the Indian equity market. Results from five stocks may not generalize to a broader universe.

Short time period: the study covers approximately three years. Momentum and mean reversion effects documented in the wider literature are often studied over much longer periods, and three years may not capture a full range of market conditions.
Concentrated portfolio: holding only two of five stocks at a time increases exposure to the specific ups and downs of individual companies, as discussed in Section 7.3.
Zero baseline transaction costs: the main results assume no cost to trade, which Section 7.2 shows matters a great deal given how often both strategies rebalanced.
Zero slippage: the baseline also assumes trades execute exactly at the recorded price, with no gap between the intended and actual trading price.
Price-index benchmark: the Nifty 50 Price Index excludes dividends, so the benchmark may understate the true passive return an investor could have earned, as discussed in Section 7.4.
No direct investor psychology data: the behavioral interpretations in Section 8 are drawn from the wider literature, not measured directly in this project.
No out-of-sample validation: all results in this paper come from the same single historical window; the strategies were not tested on a separate, unseen period.
Simple, one-month signal: only the previous month's return was tested as a signal; other lookback windows or combinations of signals were not explored.
Historical performance does not establish future performance: even a strategy that had done well in this sample would not be guaranteed to do well going forward, and a strategy that did poorly here is not guaranteed to do poorly in the future.

Each of these limitations narrows what this paper can honestly claim. Taken together, they mean this project should be read as an initial, transparent test of two simple ideas on one small dataset, not as a definitive verdict on momentum or mean reversion investing in India.

10. Future Research

Several extensions could meaningfully build on this project. Testing a larger universe, such as the Nifty 100 or Nifty 500, would reduce the concentration concern discussed in Section 7.3 and give a more representative picture of Indian large- and mid-cap behavior. Longer lookback periods, or combinations of multiple lookback windows, could be tested against the simple one-month signal used here. Alternative mean-reversion definitions, such as a statistical z-score or a moving-average-based signal, could be compared against the simple return-based version tested in this paper. Volatility filters, which might avoid trading during unusually turbulent months, and sector-neutral construction, which would prevent a strategy from accidentally concentrating in one industry, are both natural next steps.

On the cost side, a more carefully modeled transaction-cost and slippage assumption, ideally based on real brokerage data, would give a more grounded picture than the illustrative stress test used in Section 7.2. On the benchmark side, comparing against the Nifty 50 Total Return Index rather than the Price Index would give a fairer picture of passive investing, since it would include dividends. Out-of-sample testing and walk-forward testing, where a strategy is built on one period and then tested on a separate, later period it never saw during development, would give a much stronger sense of whether these patterns are reliable rather than specific to this one sample. Regime analysis, factor models, and machine-learning-based ranking methods represent more advanced directions that could build on the same underlying dataset. Finally, direct behavioral or sentiment data, such as

survey-based investor sentiment indices, could eventually allow a future project to test the behavioral explanations discussed in Section 8 more directly, rather than only as theoretical context.

All of these are proposed as future work. None of them were tested in this project, and none should be read as claims about what a more advanced version of this study would find.

11. Conclusion

This paper set out to answer one question: can simple systematic strategies based on momentum and mean reversion outperform passive investing in Indian large-cap equities? It used a five-stock universe of Reliance Industries, Tata Consultancy Services, HDFC Bank, Infosys, and ICICI Bank, tested from August 2023 to June 2026 against the Nifty 50 Price Index.

Within the selected five-stock universe and August 2023 to June 2026 sample, neither tested systematic strategy accumulated as much value as the Nifty 50 Price Index benchmark. Momentum finished above its starting capital, while the mean-reversion strategy finished approximately flat. The results therefore do not provide evidence of outperformance under this particular specification.

This does not mean that momentum or mean reversion can never work. It means that these two specific, simple, previous-month-return versions of the ideas, tested on this specific small universe, over this specific short period, with zero baseline trading costs, did not beat a passive alternative. A different universe, a different time period, a different signal, or a more realistic set of trading costs could all produce a different result, and none of those alternatives were tested here.

What this project does show is that the process of testing an investing idea honestly, with a clear methodology, careful checks against look-ahead bias, and a willingness to report an unflattering result exactly as it came out, is achievable even at a small, individual research scale. The behavioral finance framework discussed in Section 8 offers a way to think about why patterns like momentum and mean reversion might exist in the first place, but this project is a test of two mechanical rules, not a test of investor psychology itself, and the results should be read accordingly. None of the findings in this paper should be interpreted as investment advice or as a prediction of how any strategy would perform going forward.

12. References

Barberis, N., Shleifer, A., & Vishny, R. (1998). A model of investor sentiment. Journal of Financial Economics, 49(3), 307-343.

De Bondt, W. F. M., & Thaler, R. (1985). Does the stock market overreact? The Journal of Finance, 40(3), 793-805.

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. The Journal of Finance, 25(2), 383-417.

Hong, H., & Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. The Journal of Finance, 54(6), 2143-2184.

Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. The Journal of Finance, 48(1), 65-91.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. Econometrica, 47(2), 263-291.

NSE Indices Limited. (2022). Methodology document for equity indices. https://nsearchives.nseindia.com/content/indices/Method_Nifty_50.pdf

Sharpe, W. F. (1994). The Sharpe ratio. The Journal of Portfolio Management, 21(1), 49-58.

Shiller, R. J. (2003). From efficient markets theory to behavioral finance. Journal of Economic Perspectives, 17(1), 83-104.

13. Appendix

Appendix A: Formulas

$R_{i,t} = (P_{i,t} - P_{i,t-1}) / P_{i,t-1}$ [monthly return]

$M_{i,t} = (P_{i,t-1} - P_{i,t-2}) / P_{i,t-2}$ [momentum / mean-reversion signal]

$R_{p,t} = 0.5 \cdot R_{1,t} + 0.5 \cdot R_{2,t}$ [portfolio return]

$V_t = V_{t-1} \cdot (1 + R_{p,t})$ [portfolio value]

$DD_t = (V_t - \text{RunningMax}_t) / \text{RunningMax}_t$ [drawdown]

Appendix B: Monthly Stock Selections (Momentum)

Table B1. Momentum monthly selections and realized portfolio returns, October 2023 – June 2026

Month	Stock 1	Stock 2	Portfolio Return
2023-10	TCS	INFY	-3.86%
2023-11	RELIANCE	HDFCBANK	+4.74%
2023-12	INFY	HDFCBANK	+7.84%
2024-01	HDFCBANK	TCS	-6.57%
2024-02	RELIANCE	INFY	+1.59%
2024-03	TCS	RELIANCE	-1.81%
2024-04	ICICIBANK	HDFCBANK	+5.10%
2024-05	ICICIBANK	HDFCBANK	-0.21%
2024-06	HDFCBANK	INFY	+10.65%
2024-07	INFY	HDFCBANK	+7.60%
2024-08	INFY	TCS	+3.94%
2024-09	INFY	TCS	-4.88%
2024-10	HDFCBANK	ICICIBANK	+0.86%
2024-11	ICICIBANK	HDFCBANK	+2.04%
2024-12	TCS	INFY	-1.47%
2025-01	INFY	HDFCBANK	-2.10%
2025-02	RELIANCE	TCS	-10.22%
2025-03	HDFCBANK	ICICIBANK	+8.75%
2025-04	ICICIBANK	RELIANCE	+8.01%
2025-05	RELIANCE	ICICIBANK	+1.22%

Month	Stock 1	Stock 2	Portfolio Return
2025-06	INFY	ICICIBANK	+1.25%
2025-07	RELIANCE	HDFCBANK	-3.14%
2025-08	ICICIBANK	HDFCBANK	-5.31%
2025-09	TCS	RELIANCE	-2.93%
2025-10	RELIANCE	HDFCBANK	+6.40%
2025-11	RELIANCE	TCS	+4.03%
2025-12	RELIANCE	INFY	+1.86%
2026-01	INFY	TCS	+0.39%
2026-02	INFY	ICICIBANK	-9.51%
2026-03	ICICIBANK	RELIANCE	-8.07%
2026-04	RELIANCE	INFY	+0.48%
2026-05	RELIANCE	HDFCBANK	-5.59%
2026-06	ICICIBANK	INFY	-1.25%

Appendix C: Monthly Stock Selections (Mean Reversion)

Table C1. Mean-reversion monthly selections and realized portfolio returns, October 2023 – June 2026

Month	Stock 1	Stock 2	Portfolio Return
2023-10	HDFCBANK	RELIANCE	-2.85%
2023-11	TCS	ICICIBANK	+2.83%
2023-12	ICICIBANK	TCS	+7.68%
2024-01	INFY	ICICIBANK	+5.41%
2024-02	HDFCBANK	TCS	+1.64%
2024-03	HDFCBANK	INFY	-3.67%
2024-04	INFY	TCS	-3.30%
2024-05	INFY	TCS	-1.10%
2024-06	TCS	ICICIBANK	+6.68%
2024-07	TCS	ICICIBANK	+6.93%
2024-08	HDFCBANK	RELIANCE	+0.96%
2024-09	RELIANCE	HDFCBANK	+1.81%
2024-10	TCS	INFY	-6.02%
2024-11	RELIANCE	TCS	+2.31%
2024-12	RELIANCE	ICICIBANK	-3.68%
2025-01	RELIANCE	TCS	+3.18%
2025-02	HDFCBANK	ICICIBANK	-0.95%
2025-03	TCS	INFY	-1.70%
2025-04	INFY	TCS	-4.36%
2025-05	INFY	TCS	+2.96%
2025-06	TCS	HDFCBANK	+2.45%
2025-07	ICICIBANK	TCS	-4.76%

Month	Stock 1	Stock 2	Portfolio Return
2025-08	TCS	RELIANCE	-0.20%
2025-09	HDFCBANK	ICICIBANK	-1.81%
2025-10	TCS	ICICIBANK	+3.03%
2025-11	ICICIBANK	HDFCBANK	+2.64%
2025-12	HDFCBANK	TCS	+0.28%
2026-01	ICICIBANK	HDFCBANK	-2.67%
2026-02	RELIANCE	HDFCBANK	-2.29%
2026-03	INFY	TCS	-7.18%
2026-04	HDFCBANK	ICICIBANK	+5.13%
2026-05	INFY	ICICIBANK	-1.16%
2026-06	RELIANCE	TCS	-5.84%

August 2023 and September 2023 are excluded from both tables above because no valid previous-month signal existed for either month, as explained in Section 5.14.

Appendix D: Robustness Results

Full robustness tables (common active-period comparison and transaction-cost sensitivity) are reported in Tables 3 and 4 in Section 7.

Appendix E: Reproducibility and Data Integrity

No missing values, duplicate dates, or coverage gaps were found in the stock or benchmark datasets.
Every month's signal was independently re-derived and confirmed to equal the prior month's realized return, with zero look-ahead mismatches across the full sample.
Exactly two securities were selected in every active month for both strategies, with weights summing to 100 percent (50 percent plus 50 percent) in every case.
No missing return was silently treated as zero; months without a valid signal were left undefined and excluded from active trading rather than assigned a zero return.
The benchmark dataset contained 35 monthly observations with no missing values or duplicate dates.